Hindi Verbal Fluency: VFT + SpAM Mental Lexicon Analysis

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INTRODUCTION + DATASET

This study checks how Hindi speakers search words in memory using timed verbal recall and spatial similarity mapping.

- VFT captures retrieval speed and semantic switching behaviour.
- SpAM captures perceived semantic geometry in 2D space.
- Prior literature: Troyer (clustering), Dautriche (word-form), Kumar (phonological shift).
- Main goal: compare semantic vs phonetic organisation in Hindi lexical search.

Participants	35 Hindi-English bilinguals
Tasks	VFT (60 sec) + SpAM (2D arrangement)
Domains	Animals, Foods, Colours, Body-parts
Core dataset	723 Hindi/Hinglish responses
Important note	Colours has low N (n=4), mainly descriptive

RESEARCH QUESTIONS + HYPOTHESES

Research Questions (RQ1-RQ7)

- RQ1: Do productivity and retrieval speed differ across domains?
- RQ2: Which participant variables (like confidence) relate to fluency outcomes?
- RQ3: Does SpAM compactness differ across domains?
- RQ4: Do participant SpAM clusters align with semantic and phonetic models?
- RQ5: Is integrated VFT+SpAM score better than VFT-only score?
- RQ6: Does phonological similarity increase across retrieval order?
- RQ7: Does phonological similarity predict higher productivity?

Hypotheses covered (H1-H13)

H	Statement
H1	Between-cluster IRT > within-cluster IRT
H2	IRT increases with serial position
H3	Word count differs by domain
H4	IRT differs by domain
H5	Confidence positively predicts total words
H6	Confidence negatively predicts mean IRT
H7	Confidence positively predicts cluster size
H8	SpAM compactness differs by domain
H9	SpAM-semantic ARI is above chance
H10	SpAM-phonetic ARI is above chance
H11	Phonological-over-semantic trend increases over order
H12	Mean phonological similarity predicts productivity
H13	Integrated-score correlation differs from VFT-only

METHODOLOGY

Analysis pipeline used in notebook implementation:

- Data merge + filtering to Hindi/Hinglish rows.
- Compute total words, IRT, cluster size, switch count.
- Shapiro-Wilk normality checks before inferential tests.
- H1: Welch t-test for within vs between cluster IRT.
- H2/H11: mixed-effects models over serial position.
- H3/H4/H8: Kruskal-Wallis for domain-wise comparison.
- H5-H7/H12: Spearman one-tailed correlations.
- H9/H10: ARI, sign test, and permutation checks.
- H13: Steiger test for dependent correlations.

TRANSFORMERS / MODELS USED

Semantic and phonetic similarity both included as required.

Model / Method	Purpose
paraphrase-multilingual-MiniLM-L12-v2	Semantic embedding for Hindi + Hinglish words (384-d)
Phonetic key representation	Sound-similarity signal for phonetic clustering/alignment
K-Means + Agglomerative + HDBSCAN	Independent clustering and robustness check
ARI / NMI / Sign test	Quantify semantic and phonetic cluster alignment

Interpretation: Multilingual transformer gives common space for Devanagari and Roman-script Hindi words.

EMBEDDING MAP

Fig: UMAP view of embedding clusters by domain (semantic structure).

Interpretation: *Dense central groups show shared semantic neighbourhood; spread regions indicate domain-specific vocabulary pockets.*

RESULTS TABLE (RQ1-RQ7, H1-H13)

All RQ/H values below are from notebook outputs only (no hallucinated statistics).

RQ	H	Test	Main Result	Decision
Found.	H1	Welch t	Within 5841.9 vs between 10960.3 ms; t=9.2995; d=1.1206; p<0.0001	Supported
Found.	H2	MixedLM	Serial-position slope not positive in any domain (0/4 domains support)	Not supported
RQ1	H3	Kruskal	Word count by domain: H=3.2639, p=0.1956	Not supported
RQ1	H4	Kruskal	IRT by domain: H=3.9028, p=0.1421	Not supported
RQ2	H5	Spearman	Confidence vs total words: rho=-0.3951, p(one)=0.9906	Not supported
RQ2	H6	Spearman	Confidence vs mean IRT: rho=+0.2755, p(one)=0.9454	Not supported
RQ2	H7	Spearman	Confidence vs cluster size: rho=-0.2259, p(one)=0.8969	Not supported
RQ3	H8	Kruskal	SpAM compactness: H=3.4451, p=0.3280	Not supported
RQ4	H9	ARI + sign	Semantic ARI mean=0.1765; sign p<0.0001; permutation direction weak	Mixed / weak
RQ4	H10	ARI + sign	Phonetic ARI mean=0.0745; sign p=0.0662	Not supported
RQ6	H11	LME	type×position beta=-0.004144, p=0.1028	Not supported
RQ7	H12	Spearman	Phonological similarity vs productivity: rho=+0.114, p=0.2570	Not supported
RQ5	H13	Steiger Z	VFT rho=-0.461 vs integrated rho=-0.323; z=0.724, p=0.4689	No diff.

Note: Colours domain low N; interpret domain-level tests carefully.

KEY FINDINGS

Main findings (simple summary):

- Strong evidence only for semantic cluster-boundary cost (H1).
- Domain effect, confidence effect, phonological facilitation mostly not supported.
- Integrated score did not beat VFT-only score statistically (H13).

IMPORTANT FIGURE 1

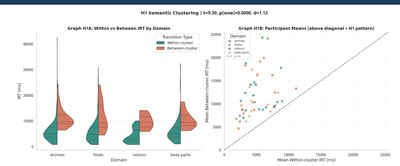

Fig: H1 cluster boundary effect (largest and clearest result).

Interpretation: *Most participant points stay above diagonal and violin distributions are clearly separated, so cluster-switch cost is strong.*

IMPORTANT FIGURE 2

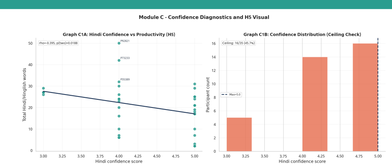

Fig: Confidence diagnostics show ceiling compression.

Interpretation: *Many participants are at high confidence band, so correlation signal becomes unstable and opposite-direction trends can appear.*

IMPORTANT FIGURE 3

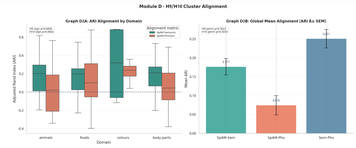

Fig: Semantic vs phonetic ARI alignment summary.

Interpretation: *Semantic ARI is generally above phonetic ARI, indicating meaning-based structure is stronger than sound-based structure here.*

SEMANTIC VS PHONETIC SUMMARY

Semantic and phonetic similarity coverage as required:

Aspect	What We Saw
Semantic	H9 shows positive semantic ARI trend, but confirmatory directional evidence is limited.
Phonetic	H10 and H12 do not support strong phonetic facilitation in this sample.
Serial trend	H11 interaction not significant (p=0.1028).
Composite score	H13: integrated score does not significantly outperform VFT-only.

IMPORTANT FIGURE 4

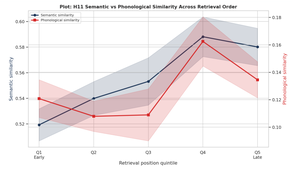

Fig: H11/H12 phonological results (no strong facilitation pattern).

Interpretation: *Curves do not show a reliable semantic-to-phonological handoff; uncertainty remains high.*

IMPORTANT FIGURE 5

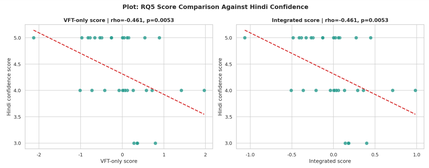

Fig: H13 comparison of VFT-only vs integrated score.

Interpretation: *Both scores show similar trend with confidence, and Steiger test says difference is not significant.*

DISCUSSION

Interpretation in simple Indian-English:

- In this sample, semantic clustering effect is very clear and strong.
- Most other hypotheses are not getting support statistically.
- This can happen due to small sample size and restricted confidence scale.
- So we should report null results honestly, and not over-claim.

Limitations:

- N=35 only; colours domain has low participation.
- Confidence self-score has ceiling effect.
- Some extended analysis blocks in notebook have inconsistent variants; we used core hypothesis outputs.

Future work:

- Larger and balanced participant pool.
- Objective Hindi proficiency test instead of only self-confidence.
- More robust phonological feature design for Hindi words.

CONCLUSION

Final conclusion: Hindi fluency retrieval shows strong semantic clustering (H1), but domain effect, confidence effect, and phonological facilitation are mostly not supported in this dataset.

VFT=Verbal Fluency Task | SpAM=Spatial Arrangement Method | IRT=Inter-Response Time | LME=Linear Mixed Effects